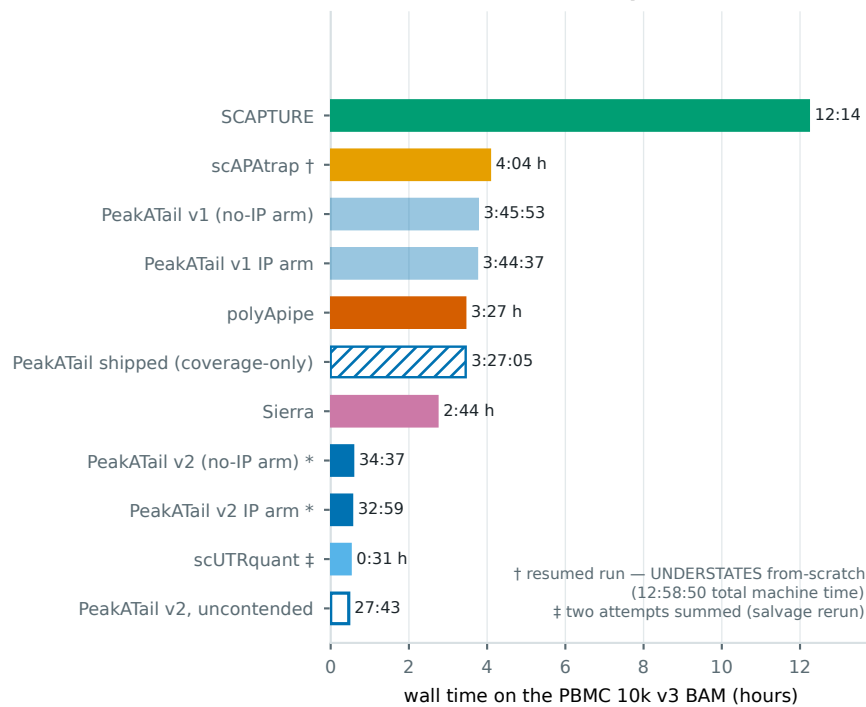
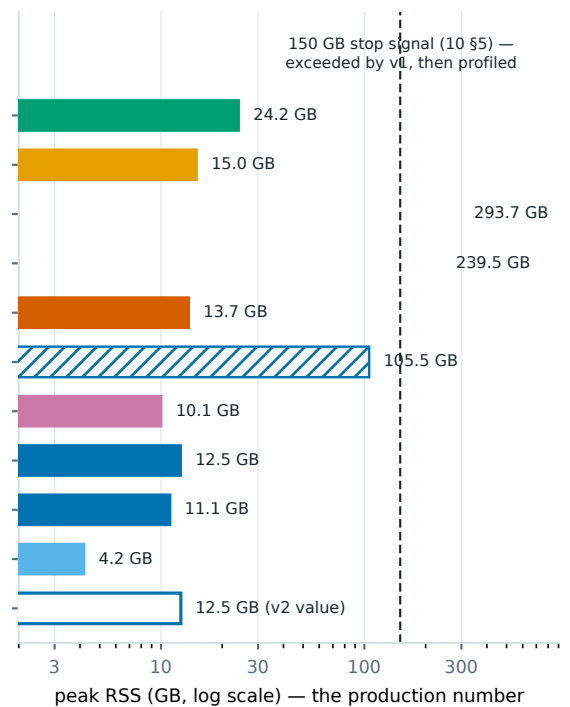


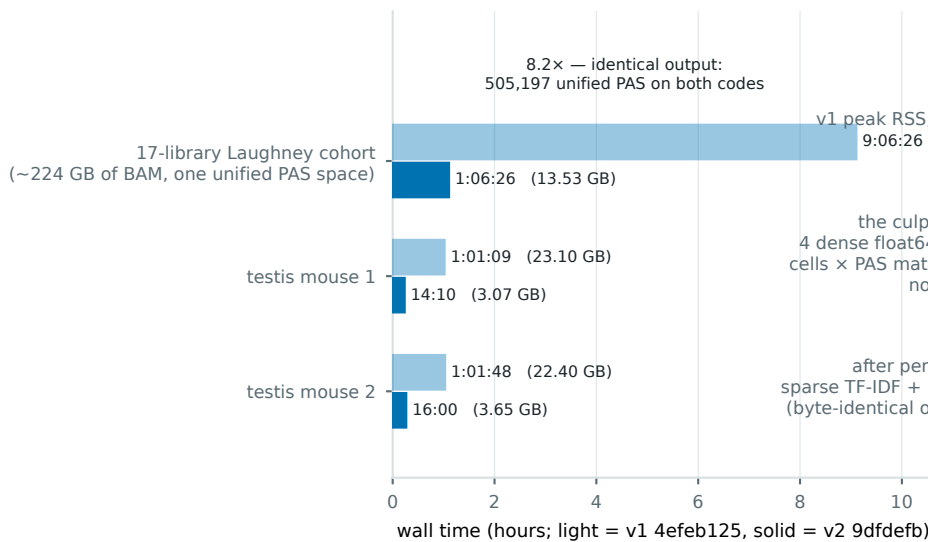
a Wall time per arm — * = concurrent, all four arms at once (never quote without this)



b Peak memory per arm: 293.7 GB → 12.53 GB (23.5×)



c The #97 fix confirmed at cohort and mouse scale (wall, with peak RSS)



d Where the memory went — profiling, not peak calling (15 \$5 addendum)

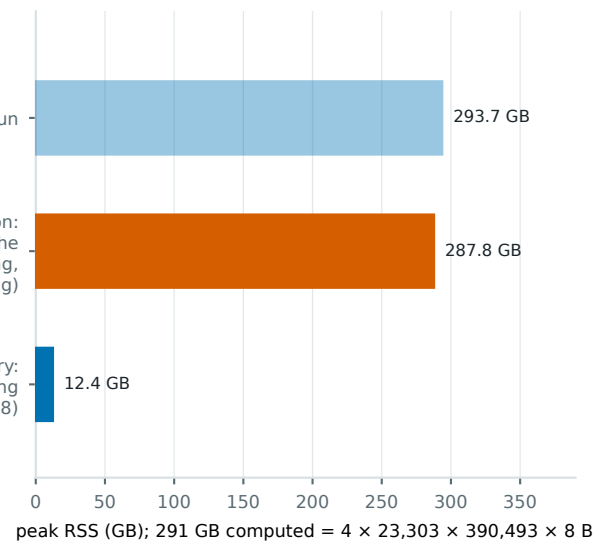


Fig S8 | Compute detail behind Fig 3d. (a, b) Every PBMC 10k v3 arm from the verified compute registry (fig3_tradeoff_compute.tsv; each bar's provenance and caveat travel in figS8_compute.tsv verbatim). Rule in force (21 \$7 item 5): never quote wall time without the concurrency disclosure — the v2 arm wall times (34:37 no-IP, 32:59 IP) were measured with all four benchmark arms running CONCURRENTLY; the uncontended single run is 27:43 — and peak RSS is the production number: 293.7 GB (v1) → 12.53 GB (v2), 23.5×. v1 exceeded the pre-registered 150 GB stop signal (10 \$5) and was profiled rather than shipped; on v1 a 10k-cell PBMC BAM needed a ≥300 GB node and used ~1.4 cores despite --threads 16 (15 \$5). Competitor caveats carried verbatim: scAPAtrop's 4:04:29 is the successful RESUMED run after an OOM-killed first attempt and UNDERSTATES a from-scratch run (12:58:50 total machine time, disclosed, not summed); scUTRquant's 30:45 sums two attempts (salvage rerun after exit 1); SCAPIRE's 12:14 sums its two mandatory stages with the annotation prebuild untimed; polyApipe and Sierra ran clean. (c) The #97 performance fix confirmed at two other scales on the same box: the 17-library Laughney cohort (~224 GB of BAM, identical YAML and flags) fell 9:06:26 → 1:06:26 (8.2×) at IDENTICAL output — 505,197 unified PAS on both codes — and the mouse arms fell 1:01:09/1:01:48 → 14:10/16:00 with peak RSS 23.1/22.4 → 3.07/3.65 GB (v1 --threads 12; v2 concurrent). (d) Why: profiling located the v1 peak in clustering (_tfidf_signac_method1), not peak calling — four dense float64 copies of the cells × PAS matrix (291 GB computed, 287.8 GB measured in isolation); the fix (sparse TF-IDF, per-(contig, strand) parallel calling, streamed CB filter) is byte-identical on all 18/18 compared outputs. Values and verbatim caveats: results/figures/manuscript/figS8_compute*.tsv.